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# Anomalous Paleomagnetic Pole from Isotopically Dated Cambro-Ordovician Intrusives in Colorado

## ABSTRACT

Paleomagnetic data from isotopically dated Cambrian and Ordovician plutons in central Colorado indicate a Cambrian-Ordovician paleopole close to the late Paleozoic pole position.

Samples were partially demagnetized by both ac and thermal techniques. Magnetic directions were essentially the same regardless of treatment. Polished section and Curie temperature analysis indicate that the principal magnetic phase is low-titanium titanomagnetite. In many samples, titanomagnetite is the only magnetic iron oxide; in some samples, hematite occurs additionally. However, those samples containing hematite show the same remanent directions as those containing only titanomagnetite. All of our studies indicate that the magnetization is thermoremanent magnetization (TRM) acquired during the Cambrian and Ordovician.

Our results suggest that either little polar wandering occurred between the early and late Paleozoic or, if substantial wandering did occur, by late Paleozoic time the paleopole has returned to a position that was nearly coincident with that of the early Paleozoic.

## INTRODUCTION

Paleomagnetic data for lower Paleozoic (Cambrian-Devonian) rocks from North America are not abundantly available (Irving, 1964; Helsley and Spall, 1966; McElhinny, 1968, 1969). The Cambrian, Ordovician, Silurian, and Devonian are represented by 6, 1, 3, and 4 paleomagnetic pole positions, respectively. (The positions of these poles and their references are given *in* Figure 6.)

Most of the lower Paleozoic data has been obtained from sedimentary rock sequences which can be dated only generally. In addition the natural remanent magnetization (NRM) of old sediments may be quite complex including components of magnetization acquired (1) during or shortly after deposition (DRM), (2) later, perhaps much later, by chemical alteration of iron oxides or iron silicates (CRM), or (3) later, by heating of the magnetic oxides (PTRM). As one might expect and as is evident from Figure 6, there is a great deal of scatter in the results from any one period. The fact that some of the lower Paleozoic paleopoles lie close to those obtained from upper Paleozoic rocks has been used by some (Irving, 1964; Creer, 1968) to indicate that those particular rocks were largely remagnetized during the late Paleozoic.

Thermoremanent magnetization (TRM), acquired by an igneous rock body as it cools through a range of blocking temperatures, is generally more reliable in determination of a paleopole position. The rock can be accurately dated radiometrically; the initial magnetization was acquired essentially at one moment and is commonly stable; later diagenetic changes occur less readily in igneous rocks (particularly intrusives) than they do in sediments. What is needed in paleomagnetic studies of the lower Paleozoic are more data from igneous rocks. In this note we report paleomagnetic data obtained from intrusive igneous rocks in Colorado which have been dated as lower Paleozoic by isotopic methods.

## GENERAL GEOLOGY

Mafic and ultramafic, commonly alkalic, plutons including gabbro, pyroxenite, dunite, diabase, anorthosite, syenite, ijolite, uncompahgrite, and lamprophyre which intrude Precambrian rocks have been recognized for a long time in the Wet Mountains (Cross, 1896; Parker and Hildebrand, 1963; Shaw and Parker, 1967), Powderhorn district (Larsen, 1942; Hedlund and Olson, U.S. Geol. Surv., open-file map, 1968), and the Black Canyon area (Hunter, 1925; Hansen, 1965) of central Colorado (Fig. 1). The alkalic complexes of the Wet Mountains and Powderhorn district are associated with carbona-

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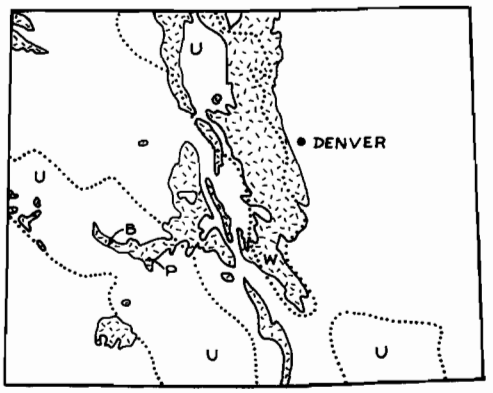



Figure 1. Index map of Colorado. Patterned areas are exposed Precambrian rocks. Areas outlined by dots and marked U are late Paleozoic uplifts. Sampled areas: B, Black Canyon; P, Powderhorn district; W, Wet Mountains.

tites (Christman and others, 1959; Olson and Wallace, 1956; Temple and Grogan, 1965). All had been considered Precambrian in age; however, radiometric dating of these plutons indicates they are Cambro-Ordovician in age (*see* Table 1).

Recent isotopic dating (Table 1) has shown that these mafic alkalic rocks fall within the Cambrian and Ordovician, as defined by Harland and others (1964, p. 241-258).

## PALEOMAGNETIC RESULTS

### Black Canyon

Samples were collected from nine sites in three diabase and syenite dikes in the Black Canyon of the Gunnison River. Sampling sites were chosen in deep road cuts or on canyon walls. A dike in the same swarm we sampled has been dated at 510 ± 60 m.y. by Rb/Sr analysis of K-feldspar (Hansen and Peterman, 1968).

The initial NRM directions for the nine sites are shown in Figure 2A. One set of nine samples was tumbled in peak ac fields up to 250 oe to remove unstable components of magnetization. After this partial ac demagnetization, the remanent directions display a tighter grouping which has shifted toward the southeast (Fig. 2B). Nine duplicate samples were thermally demagnetized by heating and cooling in field-free space (<250γ). Figure 2C shows the results after thermal demagnetization at 500° C. Some samples were taken to 550° C; this temperature was equal to or exceeded the Curie temperature of the magnetic phase of some samples and the remanent directions scattered for those samples. Where this temperature was less than the Curie temperature, the remanence remained flat lying and toward the southeast.

Both ac and thermal partial demagnetization gave grouped directions with southeast declinations and either low-angle positive or negative inclinations. The mean direction for thermally cleaned samples was essentially indentical to that for ac cleaned samples.

### Powderhorn

Four samples (two from each of two sites) were collected from a single gabbro dike about 30 mi southeast of the Black Canyon of the

TABLE 1. ISOTOPIC AGES OF LOWER PALEOZOIC PLUTONS IN CENTRAL COLORADO

| Location | Rock type | Age (m.y.) | Method | Reference |
|---|---|---|---|---|
| Black Canyon | diabase dike | 510 ± 60 | Rb-Sr | Hansen and Peterman (1968) |
| Powderhorn | syenite | 525, 583 | Pb-α | Jaffe and others (1959) |
| Wet Mountains | syenite stock McKinley Mtn. complex | 580, 601 | Pb-α | do. |
| Wet Mountains | syenite stock McKinley Mtn. complex | 500 ± 25 | K-Ar/biotite | Brock and Singewald (1968) |
| Wet Mountains | syenite dike McKinley Mtn. complex | 485 ± 24 | K-Ar/w/r† | do. |
| Wet Mountains | syenite dike McKinley Mtn. complex | 427* | K-Ar } w/r† | do. |
| Wet Mountains | syenite dike McKinley Mtn. complex | 585* | K-Ar } | do. |

*Contains devitrified glass; age may in part reflect devitrification (Brock and Singewald, 1968)
†w/r, whole rock

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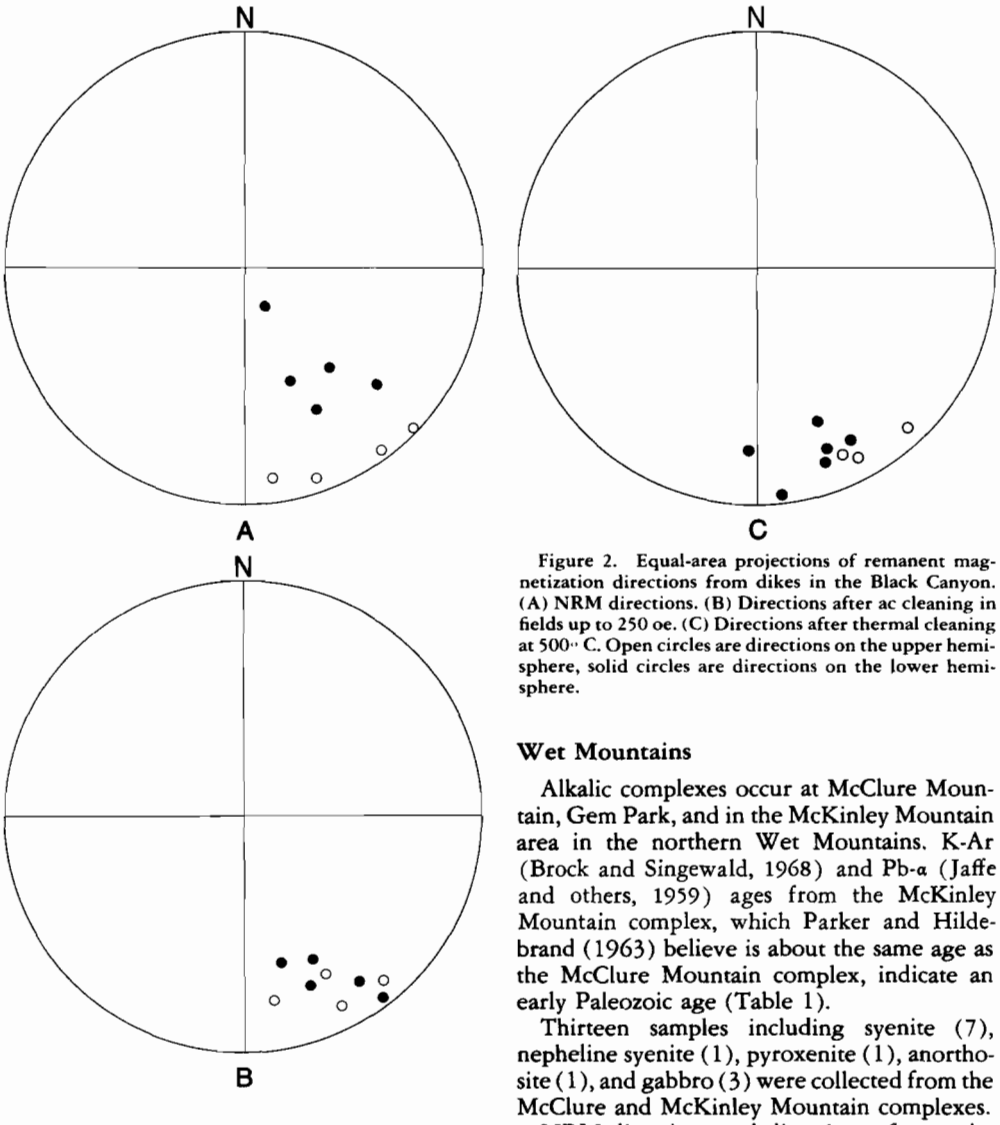



Figure 2. Equal-area projections of remanent magnetization directions from dikes in the Black Canyon. (A) NRM directions. (B) Directions after ac cleaning in fields up to 250 oe. (C) Directions after thermal cleaning at 500° C. Open circles are directions on the upper hemisphere, solid circles are directions on the lower hemisphere.

Gunnison River. The dike is a part of the Powderhorn alkalic complex which is petrographically similar to the Wet Mountain alkalic complexes. A lower Paleozoic age is suggested by Pb-α ages of 525 and 583 m.y. from zircons in Powderhorn syenite (Jaffe and others, 1959).

Two samples (one from each site) were ac demagnetized at 250 oe, and the other two were thermally demagnetized at 500° C. Figure 3 shows that all four samples have southeast declinations and low-angle positive inclinations after partial demagnetization.

## Wet Mountains

Alkalic complexes occur at McClure Mountain, Gem Park, and in the McKinley Mountain area in the northern Wet Mountains. K-Ar (Brock and Singewald, 1968) and Pb-α (Jaffe and others, 1959) ages from the McKinley Mountain complex, which Parker and Hildebrand (1963) believe is about the same age as the McClure Mountain complex, indicate an early Paleozoic age (Table 1).

Thirteen samples including syenite (7), nepheline syenite (1), pyroxenite (1), anorthosite (1), and gabbro (3) were collected from the McClure and McKinley Mountain complexes.

NRM directions and directions after ac demagnetization in fields up to 250 oe are shown on Figures 4A and 4B. After ac demagnetization, seven samples from syenite, nepheline syenite, and pyroxenite gave a grouping with a southeast declination and a low-angle positive inclination. The similarity in grouped magnetic directions after cleaning with those of the Powderhorn and Black Canyon sites lends further support to the conclusion that this is the meaningful direction of stable remanence. The remaining six samples gave scattered directions with intensities one to two orders of magnitude higher than the grouped samples. Ac cleaning

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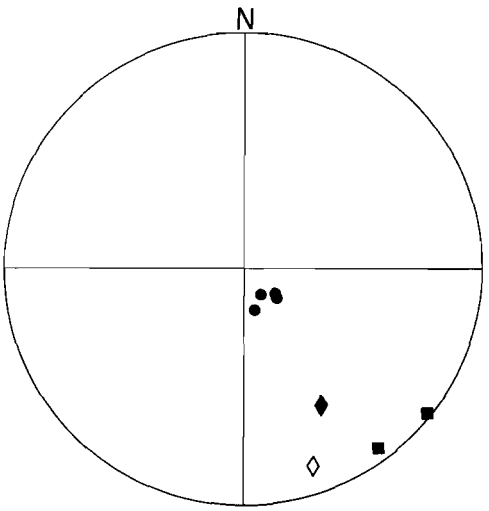




Figure 3. Equal-area projection of remanent magnetization directions from gabbro dike, Powderhorn district. Circles are NRM directions; squares are directions after ac cleaning at 250 oe; diamonds are directions after thermal cleaning at 500° C. Open symbol is direction on the upper hemisphere; solid symbols are directions on the lower hemisphere.

in fields up to 400 oe either did not shift the NRM directions, or shifted them in an erratic fashion. The collecting sites in the Wet Mountains are located on upland topography and are subject to abundant summertime thunderstorm activity. It is strongly suspected that much, if not all, of the scatter in directions is due to remagnetization by lightning discharge.

Duplicate cores were cut from six blocks—three from the grouped set (which we assume to be meaningful) and three from samples showing scattered directions and high intensities. These duplicate samples were thermally demagnetized at temperatures up to 500° C and gave essentially the same results as with ac demagnetization (Figs. 4C, 4D), except that the thermally demagnetized grouped samples have a more southerly declination than the mean for ac demagnetized samples.

## PALEOPOLE

The location of the paleopoles for remanent directions from samples from the three sites cleaned by ac and thermal demagnetization were computed separately (Table 2). These poles are shown on Figure 6 together with other paleopoles from North American rocks of early Paleozoic age.

## OPAQUE MINERALOGY

Several polished sections were made from each sampled pluton to determine if there are any relations between remanent magnetization directions and the iron-titanium oxide phases present or amount of high- or low-temperature alteration, or both, of opaque minerals. Sections were examined in reflected light under oil at magnifications up to 1000 X.

All samples show primary titanomagnetite and ilmenite in euhedral to anhedral grains ranging in diameter from $<1\mu$ to nearly 1 mm. Textures suggest that both titanomagnetite and ilmenite began to precipitate early, and continued to form throughout primary silicate crystallization. Trace amounts of primary pyrite occur in all the sections examined. Titanomagnetite and ilmenite also occur rarely as oriented exsolution (?) grains in clinopyroxene, and some magnetite occurs as an alteration product of olivine in olivine-bearing rocks.

High-temperature oxidation of titanomagnetite resulting in the development of ilmenite lamellae along (111) is present in all samples. The lamellae range from broad, widely spaced layers to very fine, closely spaced plates near the limit of microscopic resolution. The development of ilmenite lamellae in titanomagnetite apparently has been important in reducing effective grain size and increasing magnetic stability (Larson and others, 1969).

In contrast to the ubiquity of ilmenite and magnetite produced by high-temperature oxidation, the development of pseudobrookite by high-temperature oxidation and of hematite by either low- or high-temperature oxidation varies from locality to locality. Diabase and syenite dikes from the Black Canyon contain ilmenite partially altered to pseudobrookite, probably at temperatures in excess of 600° C (Lindsley, 1964). In those sections with pseudobrookite, titanomagnetite is commonly extensively altered to hematite, and finely disseminated hematite is present in both pyroboles and feldspars. We cannot determine if this hematite is exclusively of high-temperature origin, or if low-temperature hematite is also present. All Black Canyon samples, regardless of oxidation assemblage, show essentially the same remanent magnetization direction.

The Powderhorn gabbro dike contains small amounts of hematite, presumably formed at low temperatures, as an alteration product of titanomagnetite.

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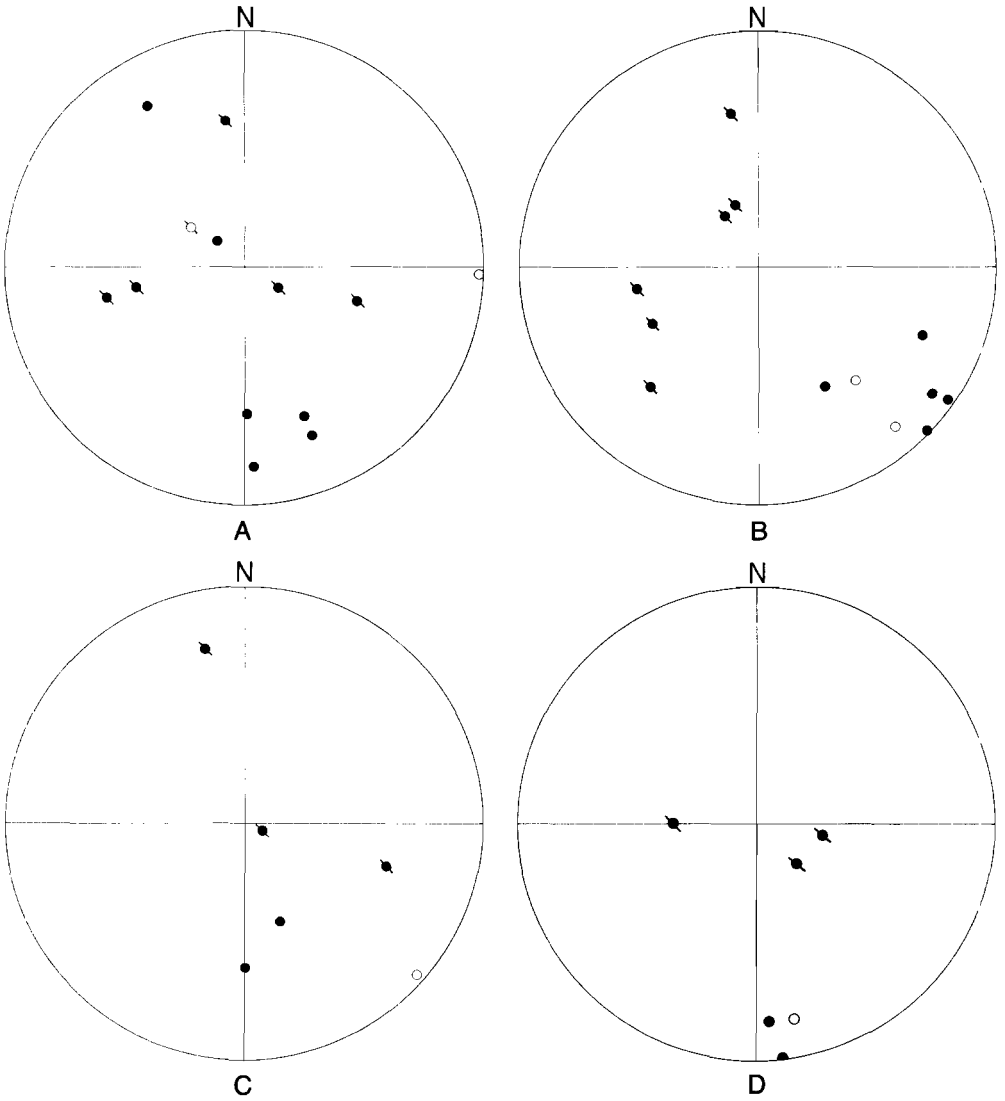




Figure 4. Equal-area projections of remanent magnetization directions from plutons in the Wet Mountains. (A) NRM directions of samples used for ac cleaning. (B) Directions after ac cleaning at 250 oe. (C) NRM directions of samples used for thermal cleaning. (D) Directions after thermal cleaning at 500° C. Open circles are directions on the upper hemisphere, solid circles are directions on the lower hemisphere. Samples omitted from statistical analysis marked with ticks.

The igneous rocks from the Wet Mountains contain high-temperature oxidation-exsolution lamellae of ilmenite in titanomagnetite, but only minor traces of hematite. No differences in opaque mineralogy between the grouped and ungrouped samples from the Wet Mountains were observed.

## CURIE TEMPERATURES

To aid in identification of magnetic phases, Curie temperature curves ($J_s$-T) were determined for selected whole rock samples heated in air at applied fields of 1000 to 6000 oe. Typical curves are presented in Figure 5.

The principal magnetic constituent is titanomagnetite with Curie temperatures that vary from 530° to 580° C, a range characteristic of titanomagnetites with low-titanium contents (Nagata, 1961). Cooling curves that nearly coincide with heating curves also indicate low-

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TABLE 2. PALEOMAGNETIC DATA AFTER PARTIAL DEMAGNETIZATION

| Cleaning method | Number of samples | Paleolatitude | Paleolongitude | $\delta M$ | $\delta P$ |
|---|---|---|---|---|---|
| ac | 18 | N. 36.6° | E. 122.3° | 6.0° | 11.9° |
| thermal | 14 | N. 44.2° | E. 99.8° | 5.4° | 10.6° |

titanium-content titanomagnetites. Also polished-section studies show that essentially all samples have undergone some high-temperature oxidation in which the ulvöspinel component of titanomagnetite is altered to ilmenite, thereby reducing the titanium content of the titanomagnetite.

The generally reversible character of the $J_s$-T curves to about 400° C indicates that maghemite is absent— a conclusion that is supported also by polished-section observation.

Although hematite is visible in some polished sections from the Black Canyon, it is evident from the lack of a high-temperature tail in the thermomagnetic curves, that even in these samples the dominant magnetic component is low-titanium titanomagnetite. This observation is supported also by the thermal demagnetization studies.

In summary, $J_s$-T analysis indicates that the principal, and in most cases the only, magnetic constituent is titanium-poor titanomagnetite.

## DISCUSSION

Paleomagnetic data may be used effectively in studies of global tectonics, continental drift, and polar wandering. Sufficient data is available from rocks of late Paleozoic and younger ages to test the hypothesis of continental drift and to draw realistic drift paths (Creer, 1967; Francheteau and Schlater, 1969; McElhinny, 1968, 1969; and Larson and LaFountain, 1970).

Relatively few paleomagnetic data are available from any part of the world for the lower Paleozoic. Figure 6 shows Paleozoic paleomagnetic data for North America. It is evident that, as yet, too few reliable data are available for accurate delineation of a possible polar-wandering trend. The paleopoles we have determined are of great interest because they do not plot near the equator, but rather, within the general grouping of late Paleozoic pole positions, as shown by Larson and LaFountain (1970). The question can be raised whether or not any significant shift in the paleomagnetic pole occurred with respect to North America during the entire Paleozoic, or whether apparent polar wandering was erratic and the pole position during the late Paleozoic nearly coincided with that of the early Paleozoic. We are currently studying the remanence of middle Paleozoic plutons in an attempt to resolve this question.

Our results may not reflect the actual Cambrian-Ordovician pole. Possibly the rocks we studied represent insufficient geologic time, and our pole represents a secular variation effect. Perhaps the plutons have been structurally rotated after crystallization, or perhaps the Colorado rocks were largely remagnetized during the late Paleozoic.

Our rocks include a variety of petrologic types and are from three relatively widely separated areas, suggesting that a considerable span of time may be represented by our samples and that they record more than a single moment or two of the past Earth's field.

The close alignment of magnetic directions from all three sampled areas suggests that little local structural rotation has occurred. Moreover, the dikes in the Black Canyon are vertical and give no indication of structural rotation. Where Paleozoic and Mesozoic strata are preserved near the sampling locations, they are generally flat-lying, or nearly so. Conceivably a

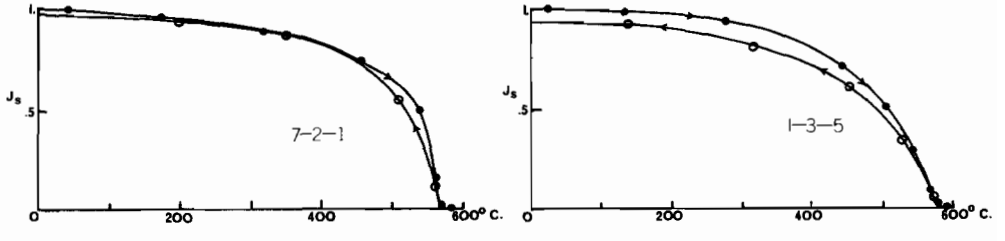



Figure 5. Curie temperature curves for two typical samples showing change in saturation magnetization ($J_s$) with temperature. Dots indicate heating curve; circles indicate cooling curve.

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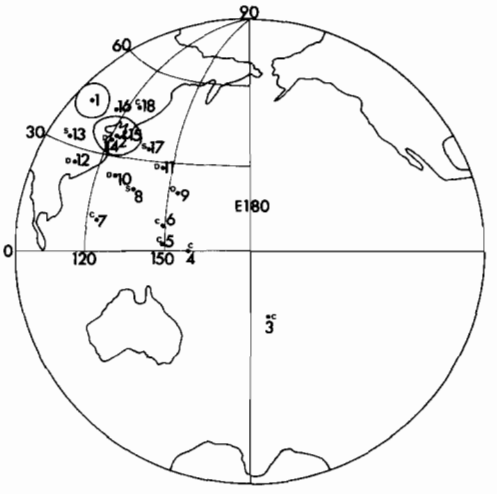




Figure 6. Equal-area equatorial projection showing the relation of our paleomagnetic pole determinations to published Paleozoic paleomagnetic data from North America. Ellipses represent 95-percent cones of confidence for paleolatitude and paleolongitude of our results. C, Cambrian; O, Ordovician; S, Silurian; D, Devonian.

(1) Cambrian to Ordovician igneous rocks from Colorado: thermal demagnetization (this paper)

(2) Cambrian to Ordovician igneous rocks from Colorado: ac demagnetization (this paper)

(3) Cambrian Ladore Formation (Irving, 1964, 2.21-2.23)

(4) Cambrian Wilberns Formation (Irving, 1964, 2.-19)

(5) Cambrian granitic igneous rocks demagnetized at 550° C (Spall, 1968)

(6) Cambrian Bradore Formation (Black, 1964)

(7) Cambrian Ratcliff Brook Formation (Black, 1964)

(8) Silurian Rosehill Formation (Irving, 1964, 4.11)

(9) Ordovician Juniata Formation (Irving, 1964, 3.-06)

(10) *Devonian Perry Lavas (Phillips and Heroy,* 1966)

(11) Devonian Clam Banks Formation (Black, 1964)

(12) Devonian Perry Lavas (Robertson and others,

(13) Silurian Bloomsburg Formation (Irving and Opdyke, 1965)

(14) Devonian sedimentary rocks with Perry Lavas (Robertson and others, 1968)

(15) Pennsylvanian mean pole position (Larson and LaFountain, 1970)

(16) Permian mean pole position (Larson and LaFountain, 1970)

(17) Silurian Clinton iron ores (Irving, 1964, 4.12)

(18) Cambrian Sawatch Formation (Howell and Martinez, 1957)

few degrees of regional rotation might have affected all three sites, but it is unlikely that rotation could exceed 5° to 10°.

The most controversial point concerning our results is the possibility of remagnetization of the Cambrian-Ordovician rocks. This problem has been faced by other workers investigating Paleozoic rocks. Spall (1968) studied isotopically dated Cambrian granitic rocks from the Wichita Mountains, Oklahoma, and concluded that both Cambrian and Carboniferous-Permian components of magnetization were present. After ac demagnetization, the rocks Spall studied displayed a direction equivalent to that of rocks of late Paleozoic age, as shown by Irving (1964). Spall concluded that the lower coercivity Cambrian component carried by magnetite was being removed during ac demagnetization. After thermal demagnetization to 550° C, Spall believed the same rocks exhibited a direction more like that from the other North American Cambrian rocks summarized by Irving (1964). Spall suggested that the "late Paleozoic" direction was the result of development of secondary hematite during the Carboniferous-Permian or possibly reflected a partial thermoremanence (PTRM) developed in hematite as a result of late Paleozoic structural uplift of the area. He remarked that the magnetic history of the Wichita Mountains may be complex.

Howell and Martinez (1957) studied the Upper Cambrian Sawatch Quartzite of north-central Colorado and reported NRM directions similar to those we obtained after partially demagnetizing our igneous rocks. Irving (1964) felt that the Sawatch direction represented a secondary magnetization, but the existence of a secondary component has never been verified by any analysis.

The Silurian Bloomsburg Formation of Pennsylvania, initially investigated by Irving and Opdyke (1965) and more recently by Roy and others (1967), has been shown to possess a secondary magnetization apparently closely related to late Paleozoic remagnetization.

A number of other rocks contain what appears to be two components of magnetization

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acquired at different times (*see*, for example, Briden, 1965). In most cases the affected rocks are sedimentary and the acquisition of the secondary magnetization seems to be associated with alteration of magnetite to hematite or a PTRM developed in secondary hematite.

In cases concerning secondary magnetization, one must usually make assumptions about the processes of primary and secondary magnetization, in what phases the primary and secondary magnetizations reside, whether the secondary phase is a single or multiple component, and the expected magnetization direction of the primary phase. Thermal and ac cleaning and fold tests are usually interpreted after considering the assumptions.

The samples in our study have been partially demagnetized, both by ac and thermal techniques. Both methods of cleaning produce very similar results (Fig. 6). This does not seem indicative of two components of magnetization—each magnetized in a different direction. Furthermore, polished-section and Curie-balance studies indicate that the only magnetic phase present in many samples is titanomagnetite; thus for these samples one cannot appeal to remagnetization of a secondary phase. The fact that samples containing both titanomagnetite and hematite also show responses to demagnetization similar to those containing titanomagnetite only indicates to us again that the observed directions are real ones acquired in Cambrian to Ordovician time during, or soon after, magmatic crystallization.

Studies by Shimizu (1960), Briden (1965), and Roy and others (1967) have indicated that rocks can develop a time-dependent viscous PTRM as a result of holding samples, for short or long periods, at elevated temperatures. It is interesting that Spall (1968) considered the secondary magnetization to be a viscous PTRM in hematite; yet the magnetite in which the primary component resided was unaffected by reheating. As pointed out by Shimizu (1960), who worked with magnetite, the effect is more pronounced the lower the bulk coercive force, and the remagnetization decays away at about one-half the rate of acquisition when the field is removed. Generally, the higher the magnetic susceptibility of a rock, the lower the bulk coercivity. For rocks of such low susceptibilities as those that we studied (*see* Table 3), it seems unlikely that they could be remagnetized completely unless the temperature was raised to essentially unrealistic values of 300° to 400° C. There is no evidence that during the Paleozoic the rocks we examined were deeply buried and at elevated temperatures.

The epizonal plutons we sampled probably crystallized and cooled within a few thousand feet of the Cambrian-Ordovician ground surface. During Cambrian through Early Pennsylvanian time the area was covered by, at most, 1000 ft of marine sedimentary rocks. During Middle Pennsylvanian through Triassic time the areas we sampled lay within linear welts which were uplifted and eroded (*see* Fig. 1). Based on the volume of derived sedimentary rocks, we estimate that perhaps 10,000 ft of material was eroded from these welts. Assuming a paleothermal gradient of 15° C per 1000 ft of depth (which is high for nonvolcanic regions), the temperature during the late Paleozoic could have been no greater than 150° C and was probably considerably less.

If remagnetization of lower Paleozoic plutons did occur, then the enclosing Precambrian crystalline rocks also should show remagnetization. Although paleomagnetic data are sparse from Precambrian rocks in Colorado, none of the results of Eggler and Larson (1968), Spall (1970, written commun.), or ourselves (current investigation) from the northern end to the south-central part of the Front Range indicates any likelihood of a late Paleozoic reheating event. The Wet Mountain Cambro-Ordovician locality is located in the southern extremity of the Front Range block, and it seems unlikely that this portion of the Front Range independently should have undergone large-scale heating in the late Paleozoic. Furthermore, the approximate concordance of K-Ar dates from biotite, crystalline whole rock, and devitrified whole rock from the Wet Mountains locality (*see* Table 1) indicates that no significant reheating took place.

## CONCLUSIONS

After consideration of our analyses, the regional geology, and the experimental results and paleomagnetic data of others, we conclude that the magnetization we have measured in samples from isotopically dated plutons is primarily a TRM that was acquired during initial cooling during Cambrian or Ordovician time, and, therefore, either little polar wandering occurred between the early and late Paleozoic, or if polar wandering occurred, the pole position in the late Paleozoic was nearly coincident with that of the early Paleozoic. It is interesting to note that the Paleozoic magnetic data from the southern-hemisphere continents indi-

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TABLE 3. SUSCEPTIBILITIES OF ROCK SAMPLES

| Sample | Location | Susceptibility e.m.u./cc |
|---|---|---|
| 1-1-2 | Black Canyon | $4.2 \times 10^{-5}$ |
| 1-3-1 | Black Canyon | $2.3 \times 10^{-5}$ |
| 1-3-3 | Black Canyon | $4.7 \times 10^{-5}$ |
| 1-4-1 | Black Canyon | $6.0 \times 10^{-5}$ |
| 1-5-1 | Black Canyon | $1.2 \times 10^{-4}$ |
| 1-6-1 | Black Canyon | $2.8 \times 10^{-5}$ |
| GU A | Powderhorn | $3.4 \times 10^{-5}$ |
| GU B | Powderhorn | $3.5 \times 10^{-5}$ |
| 7-1-1 | Wet Mountains | $5.8 \times 10^{-7}$ |
| 7-1-3 | Wet Mountains | $3.3 \times 10^{-7}$ |
| 7-1-6 | Wet Mountains | $8.1 \times 10^{-6}$ |
| 7-2-1 | Wet Mountains | $4.9 \times 10^{-5}$ |
| 7-3-1 | Wet Mountains | $1.5 \times 10^{-5}$ |
| 7-5-2 | Wet Mountains | $2.0 \times 10^{-5}$ |
| 7-7-1 | Wet Mountains | $3.6 \times 10^{-6}$ |
| 7-8-1 | Wet Mountains | $6.5 \times 10^{-5}$ |
| 7-9-1 | Wet Mountains | $4.0 \times 10^{-5}$ |
| 7-10-1 | Wet Mountains | $1.7 \times 10^{-5}$ |

cate that essentially no significant polar wandering occurred from the Cambrian through the early Devonian (McElhinny and Luck, 1970).

## ACKNOWLEDGMENTS

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## REFERENCES CITED

**Black, R. F.** Paleomagnetic support of the theory of rotation of the western part of the island of Newfoundland: Nature, Vol. 202, p. 945-948, 1964.

**Briden, J. C.** Ancient secondary magnetization in rocks: J. Geophys. Res., Vol. 70, p. 5205-5221, 1965.

**Brock, M. R.; and Singewald, Q. D.** Geologic map of the Mount Tyndall quadrangle, Custer County, Colorado: U. S. Geol. Surv., Map GQ-596, 1968.

**Christman, R. A.; Brock, M. R.; Pearson, R. C.; and Singewald, Q. D.** Geology and thorium deposits of the Wet Mountains, Colorado—a progress report: U.S. Geol. Surv., Bull. 1072-H, p. 491-535, 1959.

**Creer, K. M.** Paleozoic paleomagnetism: Nature, Vol. 219, p. 246, 1968.

**Creer, K. M.** Synthesis of world-wide paleomagnetic data: *in* Mantles of the earth and terrestrial planets (S. K. Runcorn, ed.): 351 p. Interscience Pub. Co., New York, 1967.

**Cross, Whitman.** Geology of Silver Cliff and the Rosita Hills, Colorado: U.S. Geol. Surv., 17th Ann. Rep., Part 2, p. 269-403, 1896.

**Eggler, D. H.; and Larson, E. E.** Paleomagnetic study of dated Pre-Cambrian rocks of the Front Range, Colorado-Wyoming: Roy. Astron. Soc., Quart. J., Vol. 14, p. 497-504, 1968.

**Francheteau, J.; and Schlater, J. G.** Paleomagnetism of the southern continents and plate tectonics: Earth Planet Sci. Lett., Vol. 6, p. 93-106, 1969.

**Hansen, W. R.** The Black Canyon of the Gunnison — today and yesterday: U.S. Geol. Surv., Bull. 1191, 76 p., 1965.

**Hansen, W. R.; and Peterman, Z. E.** Basement-rock geochronology of the Black Canyon of the Gunnison, Colorado: U.S. Geol. Surv., Prof. Pap. 600-C, p. C80-C90, 1968.

**Harland, W. B.; Smith, A. G.; and Wilcock, B. (eds.)** The Phanerozoic time-scale—a symposium: Geol. Soc. London, Quart. J., Vol. 120s, 458 p., 1964.

**Hedlund, D. C.; and Olson, J. C.** Geologic map of the complex of alkalic rocks at Iron Hill, Gunnison County, Colorado: U.S. Geol. Surv., open-file map, 1968.

**Helsley, C. E.; and Spall, H. R.** Compilation of paleomagnetic investigations from North America: Amer. Geophys. Union Trans., Vol. 47, p. 291-302, 1966.

**Howell, L. G.; and Maritinez, J. D.** Polar movement as indicated by rock magnetism: Geophysics, Vol. 22, p. 384-397, 1957.

**Hunter, J. F.** Pre-Cambrian rocks of the Gunnison River, Colorado: U.S. Geol. Surv., Bull. 777, 94 p., 1925.

**Irving, E.** Paleomagnetism: 399 p., John Wiley & Sons, Inc., York, 1964.

**Irving, E.; and Opdyke, N. D.** The paleomagnetism of the Bloomsburg redbeds and its possible application to the tectonic history of the Appalachians: Geophys. J., Vol. 9, p. 153-167, 1965.

**Jaffe, H. W.; Gottfried, D.; Waring, C. L.; and Worthing, H. W.** Lead-alpha age determinations of accessory minerals of igneous rocks (1953-1957): U.S. Geol. Surv., Bull. 1097-B p. 65-148, 1959.

**Larsen, E. S., Jr.** Alkalic rocks of Iron Hill, Gunnison County, Colorado: U.S. Geol. Surv., Prof. Paper 197, 64 p., 1942.

**Larson, E. E.; and LaFountain, L.** Timing of the breakup of Gondwana as determined by paleomagnetism: Earth Planet. Sci. Lett. Vol. 8, p. 341-351, 1970.

**Larson, E. E.; Ozima, Mituko; Ozima, Minoru; Nagata, T.; and Strangway, D.** Stability of remanent magnetization of igneous rocks: Roy. Astron. Soc., Quart. J., Vol. 17, p. 263-292, 1969.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010



**Lindsley, D. H.** Iron-titanium oxides: Carnegie Inst. Wash., Yr. Book, Vol. 64, p. 144-148, 1964.

**McElhinny, M. W.** Paleomagnetic directions and pole positions, VIII: Geophys. J., Vol. 15, p. 409-430, 1968.

**McElhinny, M. W.** Paleomagnetic directions and pole positions, IX: Geophys. J., Vol. 16, p. 207-224, 1969.

**McElhinny, M. W.; and Luck, G. R.** Paleomagnetism and Gondwanaland; Science, Vol. 168, p. 830-832, 1970.

**Nagata, T.** Rock magnetism: Maruzen, 350 p. Tokyo, 1961.

**Olson, J. C.; and Wallace, S. R.** Thorium and rare-earth minerals in Powderhorn district, Gunnison County, Colorado: U.S. Geol. Surv., Bull. 1027-0, p. 693-723, 1956.

**Parker, R. L.; and Hildebrand, F. A.** Preliminary report on alkalic intrusive rocks in the northern Wet Mountains, Colorado: U.S. Geol. Surv., Prof. Pap. 450-E, p. E8-E10, 1963.

**Phillips, J. C.; and Heroy, P. B.** Paleomagnetic results from the Devonian Perry lavas near Eastport, Maine: Amer. Geophys. Union, Trans. (abstr.), Vol. 47, p. 80, 1966.

**Robertson, W. A.; Roy, J. L.; and Park, J. K.** Magnetization of the Perry Formation of New Brunswick, and the rotation of Newfoundland: Can. J. Earth Sci., Vol. 5, p. 1175-1182, 1968.

**Roy, J. L.; Opdyke, N. D.; and Irving, E.** Further paleomagnetic results from the Bloomsburg Formation: J. Geophys. Res., Vol. 72, p. 5075-5086, 1967.

**Shaw, D. R.; and Parker, R. L.** Mafic-ultramafic layered intrusion at Iron Mountain, Fremont County, Colorado: U.S. Geol. Surv., Bull. 1251-A, p. A1-A28, 1967.

**Shimizu, Y.** Magnetic viscosity of magnetite: J. Geomagnet. Geoelec., Vol. 11, p. 125-138, 1960.

**Spall, Henry.** Paleomagnetism of basement granites of southern Oklahoma and its implications; Progress report: Okla. Geol. Surv., Notes, Vol. 28, No. 2, p. 65-80, 1968.

**Temple, A. K.; and Grogan, R. M.** Carbonatite and related alkalic rocks at Powderhorn, Colorado: Econ. Geol., Vol. 60, p. 672-692, 1965.

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